

AI for Science Applied to Investment Banking: An Agent-Governed, Managing-Director-in-the-Loop Architecture for M&A Target Discovery and Deal Recommendation

Alejandro Reynoso

Chief Scientist, DEFI Capital

June 10, 2026

Abstract

This paper develops an agent-governed architecture for AI-assisted investment banking and mergers-and-acquisitions advisory inspired by the *AI for Science* (AI4S) paradigm. Rather than treating a large language model as an M&A oracle or deal committee, the system places generative AI inside a closed-loop professional workflow in which candidate acquisition targets are proposed, screened for antitrust and regulatory risk, valued by deterministic engines, tested for synergy defensibility, structured against financing and board constraints, stress-tested under adverse scenarios, reviewed by a Managing Director persona, and either rejected, revised, or retained under explicit governance criteria. The architecture is implemented in a companion Colab notebook centered on Nexus Technology Corp, a mid-cap enterprise software acquirer seeking to expand into data analytics and cybersecurity through a board-approved acquisition mandate.

The paper specifies the workflow, software architecture, computational core, agent decision protocol, regulatory safeguards, LLM interfaces, stress scenarios, Managing Director review layer, and board-readiness artifacts required for an auditable AI-assisted M&A target-discovery system. The central methodological distinction is between *deal hypothesis generation* and *board-ready recommendation*. The LLM may propose target shortlists, investment theses, and strategic search directions, but it may not compute the valuation range, rewrite the synergy model, ignore financing feasibility, bypass antitrust review, or approve its own recommendation. The quantitative foundation is produced by deterministic engines: `value_target()`, `model_synergies()`, `structure_deal()`, and `score_target()`. The final continuation or stopping decision is governed by a `DealEvaluationAgent`, and no recommendation can proceed without review by an `MDReviewAgent` simulating Victoria Sands, Managing Director of M&A Advisory.

The contribution is methodological. The system is not investment advice, does not constitute a fairness opinion, does not recommend a security or transaction, and does not replace investment-banking judgment. Instead, it provides a blueprint for using AI in M&A advisory under controlled professional conditions: explicit acquirer facts, target universe definition, synthetic precedent transactions, deterministic valuation, synergy modeling, financing waterfall, EPS accretion analysis, antitrust gate, stress tests, MD-in-the-loop review, board-presentation discipline, and decision logs. The resulting architecture demonstrates how AI4S can be adapted from scientific discovery to the most numerically and strategically demanding domain of professional advisory work. Its value lies not in automating judgment away, but in making deal recommendation more systematic, transparent, auditable, and defensible before a board.

Keywords: artificial intelligence for science; investment banking; mergers and acquisitions; M&A advisory; target screening; valuation; synergy modeling; deal structuring; EPS accretion; agentic AI; board governance.

1. Introduction

A mergers-and-acquisitions mandate begins with a deceptively simple question: which company should we buy? The answer is deceptively complex. The right target must fit the acquirer’s strategic mandate, sit within the board-approved deal size range, be available at a price that can be justified by synergies, be

financeable without breaching leverage covenants, be accretive to earnings per share by year two, and clear antitrust review without material concessions. Any one of those constraints can disqualify a target that looks compelling on every other dimension. The interaction of all of them simultaneously is exactly the kind of high-dimensional search problem for which the AI4S paradigm is a useful methodological frame: a system proposes candidates, tests them against structured evidence, learns from failures, and refines its search direction. This framing follows AI4S work in which machine-learning systems accelerate candidate generation while empirical or computational tests remain the basis for acceptance [1, 2, 3].

The acquirer in this notebook is Nexus Technology Corp, a mid-cap enterprise software company providing ERP and workflow automation tools to mid-market manufacturers. Nexus seeks to acquire a data analytics or cybersecurity platform that deepens product stickiness within its existing client base, accelerates ARR growth toward \$800 million by 2027, and expands its addressable market into financial services. Nexus has \$210 million in cash, an undrawn \$150 million revolver, current leverage of approximately 1.22x EBITDA, a BBB+ credit rating, and a board-approved leverage ceiling of 3.5x EBITDA. The board has also imposed binding constraints: no deal above \$800 million without special shareholder approval, maintenance of investment-grade leverage at or below 3.5x EBITDA, EPS accretion by fiscal year two post-close, maximum dilution of fifteen percent of shares outstanding, no cross-border transaction exceeding thirty percent of combined revenue, and an integration timeline not exceeding eighteen months.

Against this mandate, the notebook defines a universe of ten candidate targets across data analytics, cybersecurity, enterprise software, and logistics technology. Each candidate has revenue, EBITDA, ARR, revenue growth, ARR growth, net debt, ownership type, employees, customer-overlap estimates, technology fit, market share, asking-price status, and strategic notes. The system must identify not merely which target is attractive, but which target can support a defensible valuation, synergy case, financing structure, EPS accretion profile, regulatory posture, and board-ready acquisition thesis.

The architecture developed here applies the AI4S grammar to M&A target discovery. A synthetic deal universe defines sector valuation benchmarks, twenty-five precedent transactions, synergy benchmarks, regulatory thresholds, and financing market conditions. An acquirer profile defines Nexus Technology Corp's strategy, balance sheet, financing capacity, board constraints, and integration limits. A deterministic analytical layer computes valuation ranges, synergy net present value, financing mix, pro forma leverage, EPS accretion or dilution, and deal score. An LLM generates target shortlists under the strategic mandate. An antitrust and regulatory gate screens each proposed target. The highest-scoring cleared candidate is stress-tested under integration failure, leverage stress, and competing bid scenarios. A simulated senior banker, Victoria Sands, reviews the recommendation for board readiness. Finally, a `DealEvaluationAgent` decides whether the system should continue searching, stop successfully, or halt after the maximum number of rounds.

The design is deliberately conservative. It does not ask the LLM to compute valuation. It does not allow a target to proceed solely because the strategic story sounds compelling. It does not allow synergy optimism to override financing constraints. It does not allow a deal to pass simply because the base case is accretive. Instead, it separates target generation, regulatory screening, valuation, synergy modeling, financing structure, stress testing, Managing Director review, and final governance into distinct modules. This separation is the central methodological contribution of the paper.

The implementation is contained in a companion Colab notebook. The executable workflow is organized into ten code cells, preceded by contextual and architecture Markdown: setup and Claude client; synthetic deal universe; acquirer profile and strategic mandate; target universe; valuation, synergy, and deal structure engines; `AcquisitionStrategyLoop`; `MDReviewAgent`; `DealEvaluationAgent`; orchestrator; and final deal-recommendation dashboard. The notebook is compact enough for classroom demonstration but rich enough to illustrate the institutional logic of governed AI in investment banking.

The paper is organized as follows. Section 2 presents the end-to-end workflow. Section 3 describes the software architecture and the separation between deterministic deal analytics, inner target-discovery loop, and outer evaluation agent. Section 4 specifies the computational core used for valuation, syn-

ergy modeling, deal structuring, accretion analysis, scoring, and stress testing. Section 5 formalizes the `DealEvaluationAgent` protocol. Section 6 describes regulatory, numerical, and governance safeguards. Section 7 explains the LLM integration. Section 8 presents the stress scenarios. Section 9 states the board-readiness standard. Section 10 discusses contributions, limitations, and future research. The appendix documents the companion notebook and includes the workflow diagram.

The intended contribution is methodological. The paper does not claim that a synthetic deal universe can substitute for real M&A diligence, fairness analysis, antitrust counsel, or investment-banking judgment. Instead, it demonstrates how AI can be placed inside a disciplined advisory workflow: one that records assumptions, computes outcomes reproducibly, challenges synergies, stress-tests financing, preserves MD judgment, and maintains an auditable decision trail.

The validation boundary is intentionally narrow. The paper evaluates internal process discipline, software separation of duties, reproducibility of arithmetic, and governance controls. It does not claim empirical predictive accuracy, superior deal selection, regulatory sufficiency, legal sufficiency, financing commitment, fairness-opinion adequacy, or readiness for unsupervised client use.

2. Workflow Overview

The workflow is a closed-loop M&A target-discovery procedure rather than a linear target-screening list. Each cycle begins with a synthetic deal universe, an acquirer mandate, and a target universe. The LLM proposes a shortlist. The system screens targets for antitrust and regulatory risk, runs deterministic valuation, synergy, financing, accretion, and scoring engines, selects the highest-scoring cleared candidate, stress-tests it, obtains structured strategic feedback, generates next-cycle seeds, and sends the result to a Managing Director review layer. The outer evaluation agent then decides whether to stop or continue.

Table 1 summarizes the ten-stage workflow implemented in the notebook.

Table 1: Ten-stage AI4S investment-banking and M&A workflow.

Stage	Description
1. Synthetic Deal Universe	Encodes sector valuation benchmarks, precedent transactions, synergy benchmarks, regulatory thresholds, and financing market conditions.
2. Acquirer Profile	Defines Nexus Technology Corp, including financials, balance-sheet capacity, strategic mandate, target sectors, deal-size range, and board constraints.
3. Target Universe	Defines ten candidate companies with financials, ownership, customer overlap, technology fit, market share, asking price, and strategic notes.
4. Target Shortlist Generation	Claude proposes a bounded shortlist of acquisition candidates with deal theses and target identifiers.
5. Regulatory Screen	Claude is prompted to perform a preliminary regulatory-risk review and classify each shortlisted target.
6. Deterministic Deal Analysis	Code computes valuation range, synergy NPV, financing mix, leverage, accretion, dilution, and deal score.
7. Candidate Selection	The system selects the highest-scoring cleared target after excluding blocked regulatory cases.
8. Stress Testing	The selected candidate is tested under integration failure, leverage stress, and competing-bid scenarios.
9. MD Review	Victoria Sands, simulated Managing Director M&A, reviews synergy defensibility, board constraints, red flags, negotiation tactics, and presentation points.
10. Evaluation Decision	<code>DealEvaluationAgent</code> decides whether to continue, stop successfully, or halt after maximum rounds.

The workflow contains two nested loops. The inner acquisition-strategy loop performs one complete M&A research cycle: shortlist generation, regulatory screening, deal analysis, best-candidate selection, stress testing, feedback, and next-cycle seed generation. The corresponding code class is `AcquisitionStrategyLoop`. The outer governance loop is implemented by `DealEvaluationAgent`. This agent governs the full process across up to three research cycles, accumulates failure context, stores rejected target identifiers, records MD verdicts, and determines whether further search is justified.

The critical control point is the regulatory-and-MD gate. A target is not advanced solely because it has a high strategic fit or attractive valuation. It must clear regulatory review, fit the board mandate, support a defensible valuation and synergy case, remain financeable within leverage constraints, become EPS accretive by year two, survive stress testing, and receive MD approval or approval with modifications. This architecture converts AI-assisted M&A from a persuasive target narrative into a governed deal-discovery process.

The workflow also imposes a hierarchy of evidence. A generated target begins as a hypothesis. It gains evidentiary status only after it is expressed as a structured target identifier with an investment thesis. It gains additional status after regulatory screening. It gains further credibility if deterministic valuation and structuring engines produce a feasible deal result. It becomes more robust if it survives integration, leverage, and competing-bid stress tests. It becomes board-plausible only after Managing Director review. Finally, it becomes a candidate recommendation only if the evaluation agent records a `STOP_SUCCESS` decision. This hierarchy is central to the AI4S mapping: the model proposes, but governance decides.

3. Software Architecture

The implementation is organized around five layers: a synthetic deal universe, an acquirer and target data layer, deterministic M&A analytics, an inner acquisition strategy loop, and an outer governance agent. This separation is not merely a software preference. It is the primary mechanism by which the notebook preserves professional accountability. Each layer has a specific responsibility and a bounded authority.

3.1. Synthetic Deal Universe

The `DEAL_UNIVERSE` object defines the M&A environment. It contains sector valuation multiples, synthetic precedent transactions, synergy benchmarks, regulatory thresholds, and financing market conditions. The sector benchmarks allow the valuation engine to use EV/EBITDA and EV/revenue approaches. The precedent deals allow the system to anchor valuation and synergy assumptions to comparable transactions. The regulatory thresholds provide a simplified basis for antitrust and market-concentration screening. The financing assumptions support cash, debt, stock, interest-cost, and accretion analysis. In the uploaded Python source, this object is encoded as `Synthetic M&A Market v1.0` for year 2024. It contains seven benchmark sectors and twenty-five precedent transactions, with financing assumptions such as a 4.5 percent risk-free rate, 1.2 percent investment-grade spread, 4.5 percent leveraged-loan spread, and a typical WACC range of 7.2 to 11.0 percent. The regulatory block includes a \$119.5 million HSR threshold, HHI reference levels of 1500 and 2500, a 30 percent market-share concern threshold, foreign-investment review, and data-privacy sensitivity for healthcare, fintech, and data analytics.

The purpose of this synthetic deal universe is not to reproduce live market data. Its methodological role is to create a controlled environment in which AI-generated target shortlists can be tested against explicit assumptions. The LLM is not permitted to invent valuation multiples, change the financing market, or override board constraints during evaluation.

3.2. Acquirer Profile and Strategic Mandate

The `ACQUIRER` object defines Nexus Technology Corp. It includes revenue, EBITDA, EBIT, net income, free cash flow, revenue growth, EBITDA margin, net debt, market capitalization, enterprise value, shares outstanding, stock price, EPS, and P/E ratio. It also records cash on hand, undrawn revolver, current leverage, maximum comfortable leverage, credit rating, target sectors, preferred deal type, maximum dilution, integration capacity, prior acquisitions, and board constraints.

The encoded financial profile is \$580 million of revenue, \$98 million of EBITDA, \$52 million of net income, \$61 million of free cash flow, \$120 million of net debt, a \$1.85 billion market capitalization, a \$1.97 billion enterprise value, 74 million shares outstanding, and a \$25.00 stock price. The code also records Nexus’s BBB+ credit rating, \$210 million of cash, and \$150 million of undrawn revolver capacity.

The strategic mandate is central. Nexus is not seeking any software asset. It is seeking a data analytics or cybersecurity platform that increases product stickiness with existing manufacturing clients, accelerates ARR growth, and opens the financial services vertical. The target universe is therefore judged not only by financial metrics but also by customer overlap, technology fit, sector adjacency, ownership availability, integration feasibility, and board acceptability.

3.3. Target Universe

The `TARGET_UNIVERSE` object defines ten candidate companies. Each candidate includes an identifier, name, sector, description, revenue, revenue growth, EBITDA, EBITDA margin, ARR, ARR growth, net debt, ownership type, employees, customer-overlap percentage, technology fit, market share, asking-price status, and strategic notes. The universe includes data-analytics platforms, cybersecurity businesses, enterprise-software assets, and logistics technology companies.

Table 2 lists the target identifiers exactly as encoded in the companion Python source. These identifiers are important because Claude may propose only targets from this fixed universe; downstream valuation, synergy, financing, and scoring functions operate on the identifier rather than on an invented company name.

Table 2: Target universe encoded in the companion Python source. Revenue and EBITDA are shown in millions of dollars.

ID	Target	Sector	Rev.	EBITDA	Fit	Overlap
TGT-001	DataPulse Analytics	Data analytics	88	18.0	high	38%
TGT-002	ShieldLayer Cybersecurity	Cybersecurity	62	8.5	medium	22%
TGT-003	Meridian Data Systems	Data analytics	145	31.0	low	8%
TGT-004	VaultGuard Security	Cybersecurity	44	4.2	very_high	44%
TGT-005	Operant Analytics	Data analytics	118	24.0	very_high	52%
TGT-006	CloudSync Enterprise	Enterprise software	76	16.0	high	29%
TGT-007	Prism Data Intelligence	Data analytics	52	7.8	medium	11%
TGT-008	NeuralOps Platform	Data analytics	38	3.2	very_high	61%
TGT-009	Fortis Cyber Defense	Cybersecurity	95	22.0	high	33%
TGT-010	Axiom Supply Chain AI	Logistics tech	68	11.5	high	41%

This data structure is important because it converts a qualitative M&A search mandate into a structured decision problem. A target can be evaluated for sector fit, growth, profitability, customer overlap, market position, ownership availability, leverage impact, and integration complexity. The LLM generates shortlists, but the downstream engines operate on target identifiers and structured fields.

3.4. Deterministic Deal Engines

The deterministic analytical layer is implemented by five pure-Python functions:

- `value_target()` for valuation range construction.
- `model_synergies()` for cost, revenue, and financing synergies.
- `structure_deal()` for financing mix, leverage, dilution, and EPS.
- `score_target()` for the normalized deal-search score.
- `full_deal_analysis()` for end-to-end target analysis.

These functions do not call Claude. They are pure arithmetic and rule-based logic.

`value_target()` computes a valuation range using sector multiples, growth adjustment, EV/EBITDA, EV/revenue, and comparable transaction logic. `model_synergies()` estimates cost, revenue, and financing synergies, then computes the net present value of expected synergy streams. `structure_deal()` applies a cash, debt, stock, and earnout financing waterfall against the acquirer's balance-sheet capacity and computes pro forma leverage and EPS accretion or dilution. `score_target()` combines the coded technology-fit mapping, customer overlap, revenue growth, EBITDA margin, synergy NPV, and a year-two accretion flag using fixed weights. It does not let Claude rewrite valuation, leverage, risk flags, or scoring weights. `full_deal_analysis()` wraps the full pipeline for a given target.

This layer is the computational center of the notebook. The LLM may generate a shortlist, but it does not directly assign valuation, accretion, leverage, or deal score. A reviewer can inspect the target data, valuation range, synergy assumptions, financing mix, leverage outcome, EPS impact, and risk flags.

3.5. AcquisitionStrategyLoop

`AcquisitionStrategyLoop` executes one research cycle. It is responsible for generating a shortlist, running the regulatory screen, analyzing all shortlisted targets, selecting the highest-scoring cleared candidate, stress-testing that candidate, obtaining strategic feedback, and generating next-cycle seeds. The output is stored in a `DealCycleResult` dataclass.

The dataclass contains cycle number, shortlist, regulatory verdicts, deal results, best target, best deal, baseline score, stress results, fragility flag, feedback verdict, and next seeds. This object functions as an audit record. It preserves both positive and negative evidence and prevents the process from relying on informal memory or hidden state.

3.6. MDReviewAgent

`MDReviewAgent` simulates Victoria Sands, a Managing Director of M&A Advisory with eighteen years of deal execution experience. Her role is not to generate new targets, but to review the AI-generated deal recommendation. She evaluates synergy defensibility, board constraints, financing structure, leverage, accretion, valuation, negotiation posture, red flags, and whether the recommendation can be presented to a board with professional credibility. Her verdict may be `APPROVE`, `APPROVE_WITH_MODS`, or `REJECT`.

This layer represents professional accountability. A deal that passes deterministic screens may still be unsuitable if the synergies are too aggressive, the integration case is weak, the board story is fragile, or the auction dynamics require a different negotiation strategy. Victoria Sands is therefore not a cosmetic reviewer. In the notebook, the evaluation agent cannot issue `STOP_SUCCESS` unless the MD review supports the recommendation.

3.7. DealEvaluationAgent

DealEvaluationAgent is the outer governor. It controls the maximum number of rounds, stores round results, stores MD reviews, accumulates failure context, records rejected target identifiers, and maintains a decision log. It sets `MAX_ROUNDS = 3`, `MIN_DEAL_SCORE = 0.60`, and `MIN_SCORE_IMPROVEMENT = 0.05`. The heuristic stop condition requires MD approval or approval with modifications, a deal score of at least 60 percent, meaningful improvement over baseline, EPS accretion by year two, and non-fragility under stress testing.

The improvement test uses `baseline_dr`, the deterministic analysis of TGT-005, Operant Analytics. In the encoded dataset this baseline has an EV midpoint of approximately \$600 million, synergy NPV of approximately \$126 million, pro forma leverage of 2.8x EBITDA, year-two EPS accretion of roughly 81.5 percent, and a deal score of about 79.1 percent.

The outer governor prevents uncontrolled search. Without this layer, a generative system could continue producing target lists until it finds one that sounds strategically compelling or produces a favorable base-case score. The evaluation agent imposes round limits, deal-score thresholds, improvement thresholds, accretion constraints, fragility controls, MD-review constraints, and failure-context memory. It converts AI-assisted M&A search into a bounded professional workflow.

4. Mathematical and Computational Core

The computational core defines the quantities the system is allowed to compute and compare. This formal layer is essential because LLMs are fluent in deal language but must not be permitted to invent valuation ranges, synergy NPVs, financing structures, accretion outcomes, or board-ready scores after observing results. The notebook therefore specifies valuation, synergy modeling, deal structuring, EPS accretion, scoring, and fragility through explicit code. The valuation and deal-structure vocabulary follows standard investment-banking and corporate-valuation practice, while remaining deliberately simplified for this prototype [4, 5, 6, 7].

4.1. Target Valuation

Let target i have revenue R_i , EBITDA E_i , and revenue growth $Growth_i$. Let m_s^{EBITDA} and m_s^{REV} denote the sector-specific EV/EBITDA and EV/revenue benchmarks for sector s . A simplified valuation estimate can be written as

$$EV_i^{EBITDA} = m_{s(i)}^{EBITDA} E_i, \quad (1)$$

and

$$EV_i^{REV} = m_{s(i)}^{REV} R_i. \quad (2)$$

The notebook uses a blended valuation approach, adjusted for growth and anchored to comparable transactions:

$$EV_i^{mid} = \alpha EV_i^{EBITDA} + (1 - \alpha) EV_i^{REV}, \quad (3)$$

with growth adjustments and comparable-transaction context applied inside the valuation engine. In the code, the sector multiple is first adjusted by

$$a_i = 1 + (Growth_i - 0.15)p_s, \quad (4)$$

where p_s is the sector growth-premium parameter and a_i is the growth-adjustment factor. When comparable precedent transactions are available, the adjusted EV/EBITDA and EV/revenue multiples are averaged with the comparable-deal means. The midpoint base value is therefore

$$B_i = \frac{1}{2} (EBITDA_i \tilde{m}_{s,i}^{EBITDA} + Revenue_i \tilde{m}_{s,i}^{REV}), \quad (5)$$

and the reported valuation range is

$$EV_i^{low} = 0.88B_i + ND_i, \quad (6)$$

$$EV_i^{mid} = B_i + ND_i, \quad (7)$$

$$EV_i^{high} = 1.15(1 + \pi_s)B_i + ND_i, \quad (8)$$

where ND_i is target net debt and π_s is the sector control premium.

The range is critical. A point estimate is insufficient for M&A governance. A board does not approve an abstract target; it approves a transaction at a price. The valuation range frames negotiation, financing feasibility, and accretion risk.

4.2. Synergy Modeling

The synergy engine decomposes total synergy into cost synergies, revenue synergies, and financing synergies. Let C_i denote annual cost synergy, Y_i denote annual revenue synergy contribution, and F_i denote financial synergy such as debt tax shield. In simplified form,

$$S_i^{annual} = C_i + Y_i + F_i. \quad (9)$$

Revenue synergy is explicitly influenced by customer overlap. In the code, for deal type d ,

$$C_i = Revenue_i c_d, \quad (10)$$

$$Y_i = (Revenue_{acq} + Revenue_i) y_d (1 + 0.50\theta_i), \quad (11)$$

$$F_i = ND_i \times 0.08 \times 0.28, \quad (12)$$

where c_d is the cost-synergy benchmark, y_d is the revenue-synergy benchmark, and θ_i is customer overlap. The realization probability is anchored to the deal-type success rate and, when available, the realized synergy rates of comparable precedent transactions. The notebook then computes synergy NPV using a five-year-style annuity with a 9 percent discount rate:

$$NPV(S_i) = S_i^{annual} q_i \frac{1 - (1 + r)^{-H_d}}{r}, \quad (13)$$

where H_d equals the benchmark realization year plus three years. This formulation is central to M&A skepticism. A deal can look attractive only if its synergy assumptions are not merely large, but defensible.

4.3. Deal Structure and Financing Waterfall

The deal-structure engine determines how the transaction is funded using cash, debt, stock, and, when applicable, an earnout. Let P_i denote purchase enterprise value. The simplified funding condition is

$$P_i = Cash_i + Debt_i + Stock_i + Earnout_i. \quad (14)$$

The notebook applies a cash-first waterfall:

$$Cash_i = \min(0.40P_i, 0.75(CashOnHand + Revolver)), \quad (15)$$

$$DebtCap_i = \max(3.5 EBITDA_{acq} - ND_{acq}, 0), \quad (16)$$

$$Debt_i = \min(P_i - Cash_i, DebtCap_i), \quad (17)$$

$$Stock_i = \max(P_i - Cash_i - Debt_i, 0). \quad (18)$$

If the target has an asking price in the dataset, the code adds an earnout equal to 10 percent of purchase value and reduces upfront cash and stock consideration by one-half of that earnout each. Pro forma net debt is

$$ND_i^{pro} = ND_{acq} + Debt_i, \quad (19)$$

and pro forma EBITDA for the leverage test is

$$EBITDA_i^{pro} = EBITDA_{acq} + EBITDA_i. \quad (20)$$

The leverage test is

$$\frac{ND_i^{pro}}{EBITDA_i^{pro}} \leq 3.5. \quad (21)$$

A target that violates the leverage ceiling cannot be treated as board-ready without modifying structure, price, timing, or financing sources.

4.4. EPS Accretion and Dilution

Let NI_{acq} denote acquirer net income. The implementation approximates target earnings contribution as $0.65 EBITDA_i$, applies interest cost of $0.055 Debt_i$, and credits 15 percent of synergy NPV in year one and 35 percent in year two. For analysis year $y \in \{1, 2\}$, pro forma EPS is

$$EPS_{i,y}^{pro} = \frac{NI_{acq} + 0.65 EBITDA_i - 0.055 Debt_i + SynCredit_{i,y}}{Shares_{acq} + \Delta Shares_i}. \quad (22)$$

EPS accretion is

$$Accretion_i = \frac{EPS_i^{pro} - EPS_{acq}}{EPS_{acq}}. \quad (23)$$

The notebook computes EPS accretion or dilution in year one and year two, with partial synergy realization in the earlier period and stronger credit in year two. The board constraint requires accretion by year two. This is a hard governance condition in the evaluation-agent heuristic.

4.5. Deal Score

The deal score combines exactly six normalized components in the uploaded code: technology fit, customer overlap, revenue growth, EBITDA margin, synergy NPV, and year-two accretion. The implemented score is

$$Score_i = 0.28Fit_i + 0.22\theta_i \quad (24)$$

$$+ 0.18 \min(Growth_i/0.40, 1) \quad (25)$$

$$+ 0.12 \min(Margin_i/0.25, 1) \quad (26)$$

$$+ 0.12 \min(NPV(S_i)/50,000,000, 1) \quad (27)$$

$$+ 0.081[AccretiveYr2_i]. \quad (28)$$

Risk flags, leverage breaches, and board-constraint violations are computed and stored separately by `full_deal_analysis()` rather than being directly subtracted inside `score_target()`. The purpose is not to claim a universal M&A score. The purpose is to rank candidate targets consistently within the synthetic mandate. The score is a search metric, not a fairness opinion.

4.6. Governance-Constrained Selection

Within a cycle, the provisional best target is selected by deal score after regulatory screening:

$$i^* = \arg \max_{i \in \mathcal{T}_{adm}} Score_i, \quad (29)$$

where $\mathcal{T}_{adm}$ denotes the set of shortlisted targets not excluded by regulatory screening. The final stopping condition is stricter:

$$\text{StopSuccess}(i^*) = 1 \quad \text{only if } Score_{i^*} \geq 0.60, \quad (30)$$

$$Score_{i^*} - Score_0 \geq 0.05, \quad (31)$$

$$AccretiveYr2(i^*), \quad (32)$$

$$\neg Fragile(i^*), \quad (33)$$

$$MDApproved(i^*). \quad (34)$$

where $Score_0$ is the baseline deal score. This formulation captures the core logic of the notebook. M&A target discovery is not unconstrained deal-score maximization. It is professionally governed strategy discovery under regulatory, valuation, synergy, financing, accretion, stress, and human-review constraints.

5. The DealEvaluationAgent Decision Protocol

The `DealEvaluationAgent` formalizes the continuation problem. In traditional M&A advisory, the decision to keep searching, refine a target, recommend a transaction, or stop a process is made through banker judgment, committee review, sponsor feedback, and client conversation. The notebook does not attempt to replace that judgment. Instead, it makes the decision protocol explicit and auditable.

5.1. Motivation

The motivation for a separate evaluation agent is governance discipline. The component that generates target shortlists should not be the same component that decides whether the deal is ready to proceed. The shortlist generator is designed to explore the target space. The evaluation agent is designed to control that exploration. This separation mirrors the professional distinction between analyst screening, associate modeling, VP review, Managing Director judgment, and board recommendation.

The agent applies five broad criteria. First, the target must be admissible from a regulatory standpoint. Second, it must produce a sufficiently strong deal score. Third, it must improve materially over the baseline. Fourth, it must satisfy EPS accretion by year two and avoid fragile stress behavior. Fifth, it must survive MD review. A target that fails any of these conditions should generally trigger another cycle or be rejected.

5.2. Two-Step Protocol

The decision protocol has two stages. The first is a deterministic heuristic. If the MD review is `REJECT`, the agent continues. If the MD approves or approves with modifications, the deal score is at least 60 percent, improvement over baseline is at least five percentage points, the deal is accretive by year two, and the deal is not fragile, the heuristic recommends `STOP_SUCCESS`. Otherwise, it recommends `CONTINUE`.

The second stage is a constrained LLM decision. The evaluation prompt receives the round number, maximum rounds, target name, deal score, baseline score, year two accretion flag, fragility status, MD verdict, MD rationale, required modifications, and heuristic recommendation. It must return a structured decision object:

```
{
  "decision": "CONTINUE | STOP_SUCCESS | STOP_MAX_ROUNDS",
  "reasoning": "short explanation grounded in metrics and MD review",
  "failure_summary": "what to fix next cycle"
}
```

The schema is intentionally narrow. The LLM may reason about the evidence, but it may not alter the contract, expand the decision vocabulary, or conceal the failure mode. If the maximum round boundary has been reached, the code prevents the model from continuing indefinitely by converting an attempted `CONTINUE` decision into `STOP_MAX_ROUNDS`.

5.3. Failure Context

A key feature of the protocol is failure-context accumulation. The current code records the best target from each continuing round, the deal score, MD verdict, required modifications, AI diagnosis, and seed names for the next cycle. That failure context is injected into the next `AcquisitionStrategyLoop`, instructing the LLM not to regenerate failed targets and to propose better alternatives.

In an AI4S framework, failure is not discarded. It becomes an experimental result. A target that fails because synergies are fragile, leverage is too high, regulatory risk is too severe, EPS accretion is not achievable, or the MD would not put the recommendation in front of a board teaches the system where not to search next. The `DealEvaluationAgent` is the mechanism that converts failed acquisition ideas into structured memory.

6. Regulatory, Numerical, and Governance Safeguards

AI-assisted investment-banking systems require safeguards because M&A advice is a high-stakes professional domain. An erroneous recommendation can destroy shareholder value, expose directors to litigation, impair credit quality, trigger regulatory delay, damage client relationships, or undermine the banker's credibility. The notebook therefore embeds safeguards at several levels: regulatory screening, structured target representation, deterministic valuation and structuring, stress testing, MD review, and stopping logic.

6.1. Regulatory Gate

The regulatory screen is mandatory. Claude is instructed to perform a preliminary regulatory-risk classification, not to act as counsel of record or provide legal advice. The screen considers market concentration, sector sensitivity, data-privacy issues, and other synthetic regulatory thresholds. Targets with unacceptable regulatory risk can be excluded before becoming the recommended candidate.

This is a critical design choice. The system should not produce a board-ready valuation thesis for a transaction that cannot plausibly clear regulatory review. In professional domains, quantification can create false legitimacy. The regulatory gate prevents a target from gaining analytical authority before it has cleared the minimum admissibility threshold.

6.2. Structured Shortlist Safeguards

The shortlist-generation prompt requires structured output. Each target recommendation must identify target IDs, thesis, rationale, expected deal type, strategic logic, and constraints. This structure converts open-ended M&A language into an inspectable object. A recommendation that says “buy a cybersecurity company” is not testable. A recommendation that identifies TGT-004 or TGT-005, states the rationale, and submits the target to deterministic analysis is testable.

Structured representation reduces ambiguity. It also prevents the model from smuggling unsupported companies into the process. The target universe is fixed, and the code analyzes only recognized target identifiers.

6.3. Numerical Safeguards

The deterministic deal engines protect against hallucinated outcomes. They compute valuation range, synergy NPV, financing mix, leverage, accretion, dilution, deal score, and risk flags from encoded assumptions and target data. This decomposition makes the result inspectable.

A production version would require additional controls, including live market data, quality-of-earnings diligence, tax structuring, purchase accounting, working-capital normalization, debt capacity analysis, antitrust counsel review, fairness committee process, board minutes, and independent model validation. The notebook demonstrates the architecture rather than a production fairness opinion or investment-banking recommendation.

6.4. Synergy-Skepticism Safeguards

The most common failure mode in M&A is not merely overpaying for the wrong target. It is paying a correct-looking price for a target whose synergies never materialize at the assumed rate. The system addresses this by decomposing synergies into cost, revenue, and financial components; anchoring synergy logic to precedent and sector assumptions; stress-testing under an integration failure scenario; and requiring Victoria Sands to explicitly scrutinize synergy defensibility before approval.

This is essential because synergy optimism can make nearly any acquisition look attractive. A governed M&A system must make synergy assumptions visible, challengeable, and stressable.

6.5. MD-Review Safeguards

The MDReviewAgent models senior banker skepticism. Victoria Sands reviews not only the deal score but also whether the synergy case is defensible, whether financing is feasible, whether board constraints are met, whether red flags are adequately disclosed, and what negotiation tactics are appropriate. Her output includes required modifications, board-presentation points, negotiation recommendations, and red flags.

This layer is essential because the real audience is the board. A model may produce an attractive base-case score, but a Managing Director must decide whether the recommendation could survive adversarial questioning by directors, general counsel, CFO, and possibly shareholders.

6.6. Stopping Safeguards

The maximum number of rounds prevents uncontrolled search. The minimum deal score threshold prevents the system from stopping on a weak candidate. The minimum improvement threshold prevents the system from recommending a target that is not meaningfully better than baseline. The EPS-accretion condition enforces a board mandate. The fragility rule prevents a deal from advancing if it collapses under adverse stress. The MD-review condition prevents model confidence from replacing professional accountability. Together, these controls convert AI-assisted M&A screening into governed deal exploration. This separation among model output, deterministic calculation, and accountable human review is consistent with model-risk and AI-risk-management principles [8, 9, 10].

7. LLM Integration

The LLM is integrated as a constrained reasoning and generation component. It is not treated as a valuation model, fairness committee, antitrust counsel of record, financing source, or final decision maker. Its primary functions are target-shortlist generation, regulatory screening, strategic feedback, next-cycle seed generation, MD-style review, and evaluation-agent reasoning. Each function is governed by a prompt contract and expects structured output.

The notebook uses Anthropic Claude through a centralized function. The model identifier is stored in MODEL:

```
MODEL = 'claude-haiku-4-5-20251001'
```

The wrapper function is:

```
ask_claude(system, user, max_tokens=1200)
```

This function sends system and user prompts to the model and returns text. The setup installs the Anthropic SDK, retrieves the API key from Colab userdata, and constructs an Anthropic client. A separate parsing utility removes common Markdown code fences and parses JSON. This design is important because LLM outputs can be semantically rich but structurally unstable. The surrounding code enforces the interface.

The implementation uses six prompt families:

1. **Shortlist generation.** Claude proposes target shortlists and deal theses using target identifiers from the approved universe.
2. **Regulatory screening.** Claude performs a preliminary regulatory-risk classification before targets proceed to final ranking.
3. **Strategic feedback.** Claude evaluates the selected deal after deterministic analysis and stress testing, identifying vulnerabilities and improvements.
4. **Next-cycle seed generation.** Claude proposes two refined search directions based on feedback and rejected target names.
5. **MD review.** Claude simulates Victoria Sands, Managing Director M&A, and returns a professional verdict, modifications, negotiation recommendations, red flags, and board-presentation points.

6. **Evaluation-agent decision.** Claude helps the outer governor decide whether to continue, stop successfully, or halt at maximum rounds.

The most important design principle is that LLM judgment is never allowed to replace deal governance. The LLM may propose targets and produce structured commentary, but it does not compute valuation independently, does not bypass the regulatory gate, does not override board constraints, does not replace the deal-structure engine, and does not substitute for MD review. Stronger models may improve the quality of target hypotheses, but they do not remove the need for deterministic analytics, stress testing, banker review, and audit logs.

8. Stress Testing Scenarios

Stress testing asks whether a candidate deal remains credible when important assumptions deteriorate. A deal that looks attractive under the base case may become weak if synergies fail, financing costs rise, leverage capacity tightens, or a competing bidder forces an increased purchase price. The notebook therefore subjects the selected candidate in each cycle to three stress scenarios.

The scenarios are illustrative rather than exhaustive. Their purpose is to model the governance principle that a recommendation should not be judged only in the base case. A production system would include downside cases for revenue retention, customer churn, product integration, employee attrition, regulatory delay, purchase-accounting adjustments, working-capital leakage, tax leakage, interest-rate shocks, credit-market closure, and auction-process escalation.

Scenario A — Integration Failure. The notebook reduces synergy realization to 30 percent of the projected case. This scenario tests the most common M&A failure mode: a deal that appears defensible only because management assumes synergies that cannot be captured in practice. If the deal score or accretion profile depends too heavily on full synergy realization, the target should be treated as fragile.

Scenario B — Leverage Stress. The system tests the deal under tighter financing conditions. In the uploaded code, this scenario applies a 10 percent purchase-price uplift, recomputes the deal structure, and then subtracts four percentage points from year-two EPS accretion as a financing-cost shock proxy. This scenario tests whether the transaction remains financeable while preserving investment-grade comfort and the board's maximum leverage threshold.

Scenario C — Competing Bid. The system increases the purchase price by 20 percent to reflect auction tension or a competing strategic bidder. This scenario tests whether the target remains attractive if Nexus must pay above the initial valuation midpoint. It also forces a negotiation question: at what price should Nexus walk away?

A deal is classified as fragile in the code if the integration-failure case is not accretive by year two, the leverage-stress case exceeds Nexus's 3.5x EBITDA comfort ceiling, or the competing-bid case is not accretive by year two. Fragility does not mean that the target has no strategic value. It means that the deal should not proceed without a modified valuation, stronger diligence, renegotiated price, revised financing structure, or explicit board disclosure. In the AI4S loop, fragility becomes structured feedback for the next cycle.

9. Board-Readiness Standard

The real audience for an M&A recommendation is not the model, the notebook, or the analyst who produced the screen. The real audience is the board of directors. A board-ready recommendation must survive questions from directors, the CFO, general counsel, outside counsel, financing providers, and sometimes shareholders. The relevant question is not whether the AI system can produce a plausible acquisition thesis. The relevant

question is whether the workflow creates a controlled, reviewable, and challengeable process for producing a recommendation that a senior banker would be willing to present.

The board-readiness standard separates three approval objects. The first is the *methodology*: whether the architecture cleanly separates shortlist generation, regulatory screening, valuation, synergy modeling, deal structuring, stress testing, MD review, and final governance. The second is the *model*: whether assumptions, code, prompts, stochastic or deterministic outputs, and decision thresholds are versioned, validated, and reproducible. The third is the *transaction recommendation*: whether the specific deal is strategically compelling, financeable, accretive, regulatory-feasible, synergy-defensible, and suitable for board action.

Table 3 summarizes the board-level challenge process.

Table 3: Board-readiness challenge framework for the M&A prototype.

Board question	Notebook evidence	Production control required
Strategic fit	Target sector, customer overlap, technology fit, strategic notes, and acquirer mandate.	Management interviews, product diligence, customer calls, competitive analysis, and strategic alternatives review.
Valuation defensibility	Sector multiples, precedent deals, EV/EBITDA, EV/revenue, growth adjustment, and valuation range.	Live market data, banker fairness work, discounted cash flow, trading comps, transaction comps, and sensitivity analysis.
Synergy credibility	Cost, revenue, and financial synergy components plus synergy NPV and stress test.	Functional diligence, integration plan, synergy owner assignment, timing schedule, capture probability, and synergy tracking.
Financing feasibility	Cash/debt/stock waterfall, leverage test, dilution cap, and accretion analysis.	Financing commitments, rating-agency dialogue, debt-market check, treasury review, and legal documentation.
Regulatory feasibility	Antitrust and regulatory screen using synthetic thresholds.	Counsel-led antitrust analysis, HSR or equivalent review, data-privacy analysis, market-definition work, and remedies assessment.
Board accountability	MD review, board presentation points, negotiation tactics, red flags, and decision log.	Formal board materials, banker sign-off, fairness committee process, minutes, conflicts review, and legal opinions.

This framing clarifies what the board should challenge. First, the deal score is useful for search but is not a fairness opinion. Second, the synergy model is a structured estimate, not proof of capture. Third, the regulatory screen is pedagogical and cannot replace counsel. Fourth, accretion analysis is sensitive to financing cost, share issuance, price, and synergy timing. Fifth, stress testing is illustrative and must be expanded before any live transaction recommendation.

A board or investment committee could defensibly approve the architecture for controlled research, analyst training, or supervised M&A idea generation if seven conditions are met. The system must maintain immutable prompt, response, model-version, code, and assumption logs; use only approved data sources; require banker approval before client or board communication; prohibit the LLM from changing valuation or financing assumptions; disclose the synthetic nature of the deal universe; escalate all fragile, regulatory-risk, leverage-sensitive, or synergy-dependent recommendations; and document why each continuing or stopping decision is consistent with the acquirer's mandate.

The committee should not approve automated deployment from the notebook as written. The correct

approval boundary is narrower: the notebook demonstrates a credible governance architecture that can be advanced to model validation, banker workflow integration, and controlled internal use. It is strongest where it makes judgment explicit and most incomplete where real deals are hardest: diligence, negotiation, financing commitments, tax structuring, regulatory review, management quality, cultural integration, and board accountability.

10. Discussion and Limitations

The architecture should be evaluated as a governance design, not as a claim of real-world M&A completeness. Its primary strength is the explicit connection between generative AI, deterministic valuation, synergy modeling, deal structuring, regulatory screening, stress analysis, MD review, decision logging, and board-readiness challenge. Its primary limitation is that the deal universe, precedent transactions, financing conditions, regulatory thresholds, and target facts are synthetic and simplified.

10.1. Contributions

The first contribution is a formal mapping from AI4S to investment banking and M&A advisory. The paper shows how observation, representation, target generation, regulatory screening, valuation, synergy modeling, stress testing, feedback, and revision can be implemented as a professional deal-advisory workflow. The second contribution is the separation between deterministic M&A engines and LLM-generated deal hypotheses. The third contribution is the separation between `AcquisitionStrategyLoop` and `DealEvaluationAgent`. This prevents the component that generates and tests targets from also owning the final governance decision. The fourth contribution is the explicit Managing-Director-in-the-loop layer represented by `MDReviewAgent`. The fifth contribution is the insistence that synergy optimism be stress-tested and reviewed before any recommendation is treated as board-ready.

10.2. Limitations

The most important limitation is that the deal universe is synthetic. Real M&A advisory would require live financial data, management guidance, quality of earnings, customer diligence, product diligence, market diligence, legal diligence, tax diligence, financing commitments, regulatory analysis, board process, conflicts review, and fairness work.

A second limitation is that the notebook is not empirically validated against historical M&A outcomes. It demonstrates internal governance logic and calculation reproducibility, but it does not estimate the probability of deal completion, post-close value creation, regulatory clearance, or shareholder return.

A third limitation is that the Managing Director is simulated by an LLM. This is useful for prototyping, but it is not a substitute for senior banker review. In a real investment-banking practice, the system should escalate candidate recommendations to a responsible banker before any client or board communication.

A fourth limitation is that valuation is simplified. Real valuation work would include trading comparables, precedent transactions, discounted cash flow, leveraged buyout analysis, sum-of-the-parts analysis where relevant, premium paid analysis, sensitivity tables, and fairness ranges. The notebook's valuation engine demonstrates structure rather than live fairness work.

A fifth limitation is that the synergy and integration models are simplified. Real synergy capture depends on functional ownership, integration timelines, retention plans, customer churn, product roadmap, cultural fit, systems integration, one-time costs, and execution risk. The notebook captures these ideas through structured fields and stress scenarios, but not through a full post-merger integration model.

A sixth limitation is stochasticity in LLM-generated shortlists. Different model calls may produce different candidates and theses. Diversity can improve search, but it also requires logging, versioning, reproducibility controls, and prompt governance. A production system would need to store prompts, responses, model versions, deal-universe versions, target-data versions, and calculation hashes.

10.3. Future Work

Future work should proceed along five lines. First, the synthetic target universe should be replaced with a live deal screener connected to actual financial, ownership, funding, and sector data. Second, the valuation engine should be expanded to include trading comps, precedent comps, DCF, LBO, premium-paid analysis, and sensitivity tables. Third, the synergy model should be linked to post-merger integration milestones, one-time costs, retention assumptions, revenue-dis-synergy risk, and realized synergy tracking. Fourth, the regulatory gate should be strengthened with counsel-reviewed antitrust, data-privacy, and sector-specific review workflows. Fifth, the dashboard should export model cards, risk registers, board memos, negotiation playbooks, reproducibility manifests, fairness-process checklists, and audit bundles.

A particularly important extension is a post-close learning loop. Every acquisition produces evidence that should inform the next one: realized synergy capture, integration duration, employee retention, customer churn, revenue retention, product integration success, and actual EPS impact. A mature AI4S M&A system should feed those post-close outcomes back into future synergy benchmarks and stress assumptions.

11. Conclusion

This paper has presented an AI4S architecture for agent-governed investment banking and M&A target discovery. The system combines a synthetic deal universe, an acquirer profile, a structured target universe, deterministic valuation, synergy modeling, deal structuring, LLM-generated shortlist search, mandatory regulatory screening, stress testing, Managing Director review, board-presentation artifacts, and a supervisory evaluation agent. Its central design principle is that AI creativity must be embedded inside professional deal governance.

The architecture is not an M&A chatbot. It is a controlled discovery loop. The LLM proposes targets, but does not authorize them. The regulatory gate excludes transactions that should not be promoted as legitimate options. The valuation engine computes ranges rather than accepting narrative price claims. The synergy engine decomposes value creation into cost, revenue, and financial components. The deal-structure engine tests financing feasibility, leverage, stock issuance, and EPS accretion. The stress tests expose fragility under integration failure, leverage pressure, and auction tension. The MD review introduces commercial skepticism and board-readiness judgment. The evaluation agent coordinates these layers and records why the process continues, stops successfully, or halts at the maximum-round boundary.

The deeper point is that M&A has always been a propose-test-learn process. Bankers generate target lists, test strategic fit, build valuation cases, challenge synergies, structure financing, negotiate price, update the thesis after diligence, and revise advice as uncertainty resolves. What AI4S adds is not a replacement for that professional process, but a disciplined way to make it more systematic. The model can search more broadly, evaluate alternatives in parallel, preserve negative evidence, and make failure analysis explicit rather than tacit.

The companion notebook makes this claim operational. It shows how one M&A mandate can be converted into a structured AI4S loop: propose, screen, value, structure, stress-test, review, decide, and iterate. The system cannot replace the judgment that comes from knowing the client, reading the market, managing a process, negotiating with counterparties, and standing in front of a board. Those judgments belong to the human banker. What the system can do is organize the search space, evaluate candidates

consistently, preserve failure context, and make recommendations more transparent. Its value lies not in replacing investment-banking judgment, but in making that judgment more informed, disciplined, auditable, and board-defensible.

Appendix A. Companion Computational Notebook

The paper is accompanied by one computational notebook:

AI4S Investment Banking Notebook — companion Colab notebook

The linked notebook corresponds to the uploaded Colab-exported Python source:

```
uploads/topic_730_ai4s_in_investment_banking_and_mergers_and_acquisitions.py
```

The export is intended to be run in Google Colab and therefore retains notebook setup conventions such as package installation and Colab userdata access. The notebook implements the architecture described in this paper. It contains a complete, self-contained demonstration of a governed agentic M&A target-discovery workflow. The executable implementation is organized into ten code cells, accompanied by contextual, architecture, reference, and conclusion Markdown:

1. Setup, dependencies, Claude client, and parsing utilities.
2. Synthetic deal universe: sector valuation benchmarks, precedent transactions, regulatory thresholds, and financing conditions.
3. Acquirer profile and strategic mandate for Nexus Technology Corp.
4. Target universe with ten candidate companies.
5. Valuation, synergy, and deal-structure engines.
6. AcquisitionStrategyLoop.
7. MDReviewAgent.
8. DealEvaluationAgent.
9. Orchestrator and execution entry point.
10. Final deal-recommendation dashboard and visual audit trail.

The notebook should be interpreted as a methodological prototype. It does not represent investment advice, does not constitute a fairness opinion, does not recommend any security or transaction, and does not create an advisor-client relationship. Its purpose is to show how AI-assisted investment banking can be governed through explicit acquirer facts, synthetic market assumptions, structured target data, deterministic valuation, synergy modeling, financing analysis, regulatory gates, stress tests, MD review, board-presentation discipline, and decision logs.

Appendix B. Workflow Diagram

Figure B.1 provides the workflow diagram corresponding to the agent-governed AI4S investment-banking and M&A architecture described in Section 2. It summarizes the movement from acquirer mandate and target universe to shortlist generation, regulatory screening, deterministic valuation, synergy modeling, deal structuring, stress testing, Managing Director review, and evaluation-agent decision.

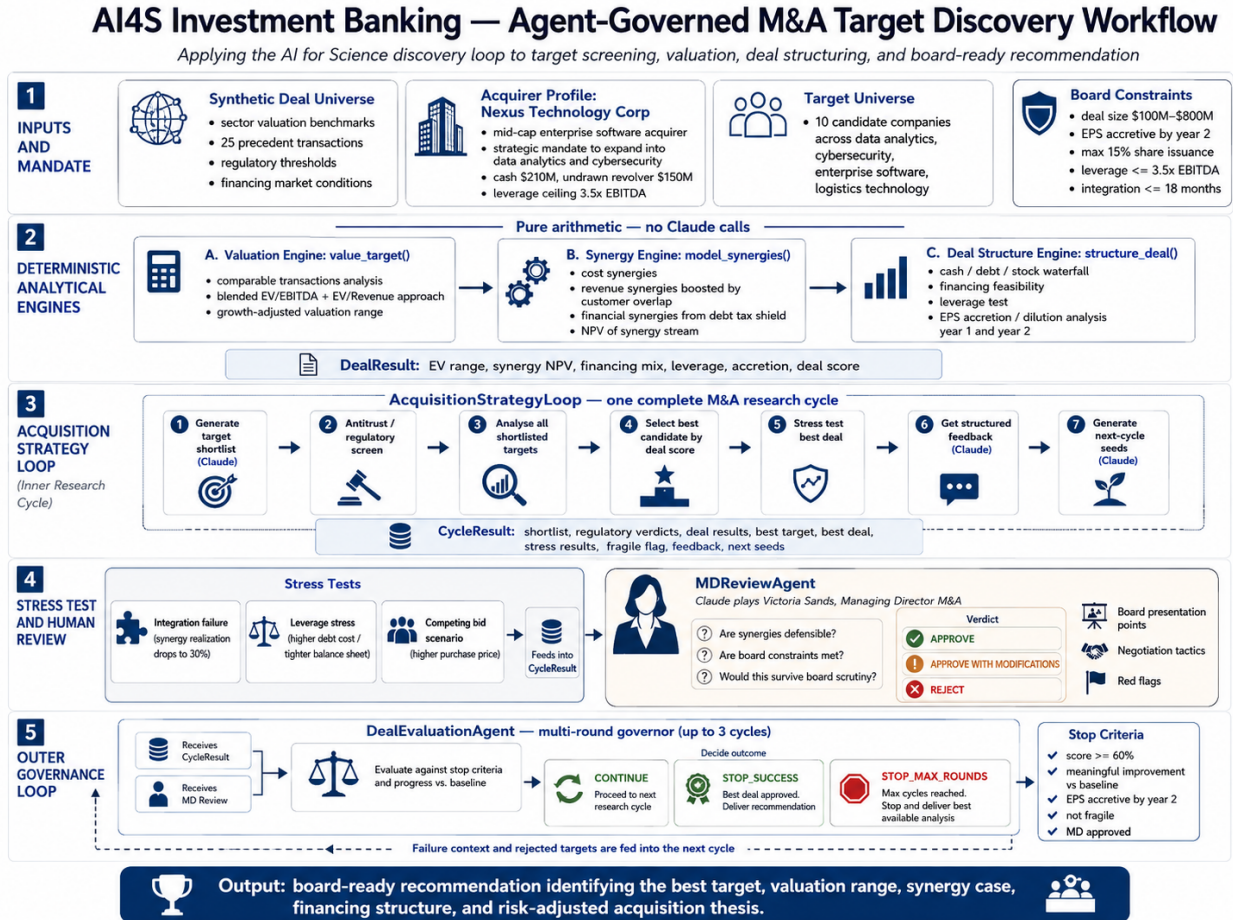


Figure B.1: Workflow diagram for the agent-governed AI4S investment-banking and M&A architecture.

References

- [1] H. Wang *et al.*, “Scientific discovery in the age of artificial intelligence,” *Nature*, vol. 620, pp. 47–60, 2023.
- [2] J. Jumper *et al.*, “Highly accurate protein structure prediction with AlphaFold,” *Nature*, vol. 596, pp. 583–589, 2021.
- [3] B. Burger *et al.*, “A mobile robotic chemist,” *Nature*, vol. 583, pp. 237–241, 2020.
- [4] A. Damodaran, *Investment Valuation: Tools and Techniques for Determining the Value of Any Asset*. Wiley, 2012.
- [5] J. Rosenbaum and J. Pearl, *Investment Banking: Valuation, Leveraged Buyouts, and Mergers and Acquisitions*. Wiley, 2013.
- [6] T. Koller, M. Goedhart, and D. Wessels, *Valuation: Measuring and Managing the Value of Companies*. Wiley, 2020.
- [7] R. F. Bruner, *Applied Mergers and Acquisitions*. Wiley, 2004.
- [8] Board of Governors of the Federal Reserve System and Office of the Comptroller of the Currency, *Supervisory Guidance on Model Risk Management*. SR 11-7 / OCC 2011-12, 2011.
- [9] National Institute of Standards and Technology, *Artificial Intelligence Risk Management Framework*. NIST AI 100-1, 2023.
- [10] S. Russell, *Human Compatible: Artificial Intelligence and the Problem of Control*. Penguin, 2020.